Wi-Fi 7 At Scale

Wes Purvis

WLPC Prague – October 2026

Notable Trend

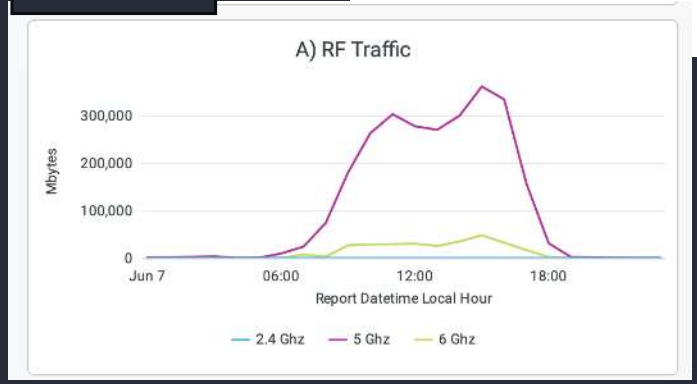
Wi-Fi Usage is Increasing

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2023

3,846.0

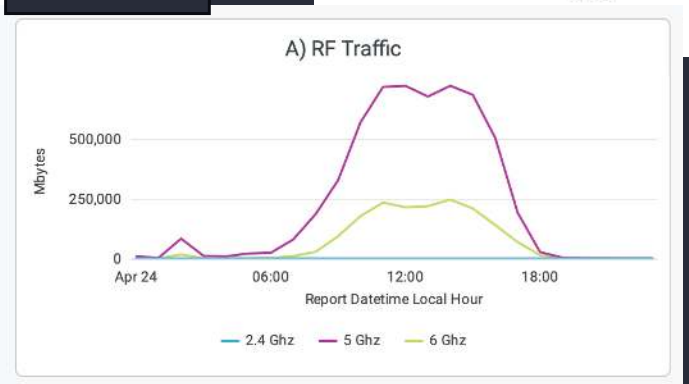
GBs



2024

6,895.0

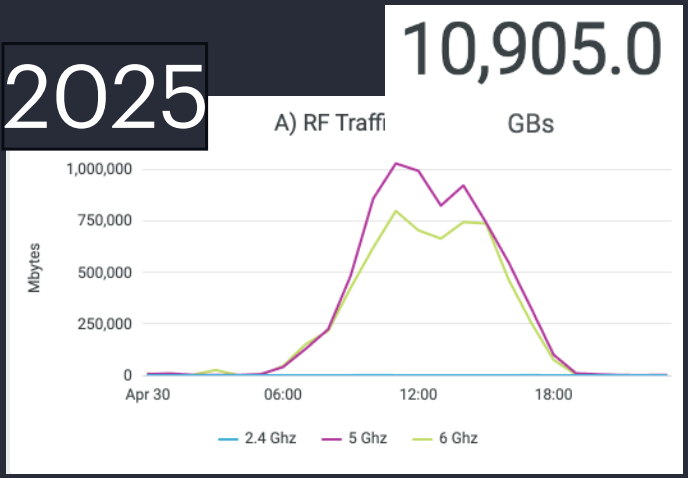
GBs



2025

10,905.0

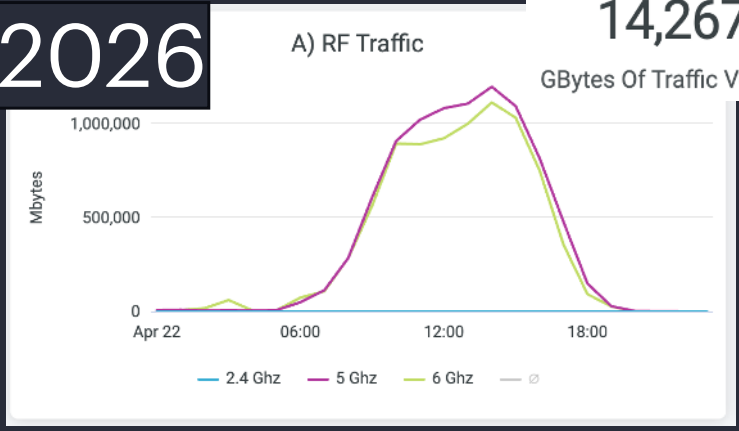
GBs



2026

14,267

GBytes Of Traffic Volume



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Solving the insatiable need for more Capacity

More Spectrum

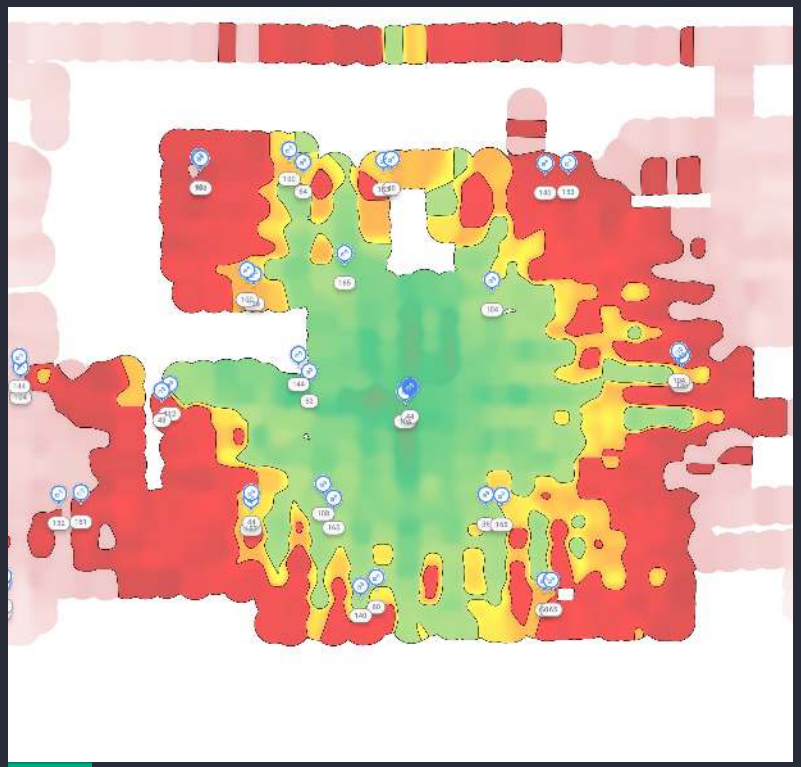
More Efficient Standards

More Efficient Designs

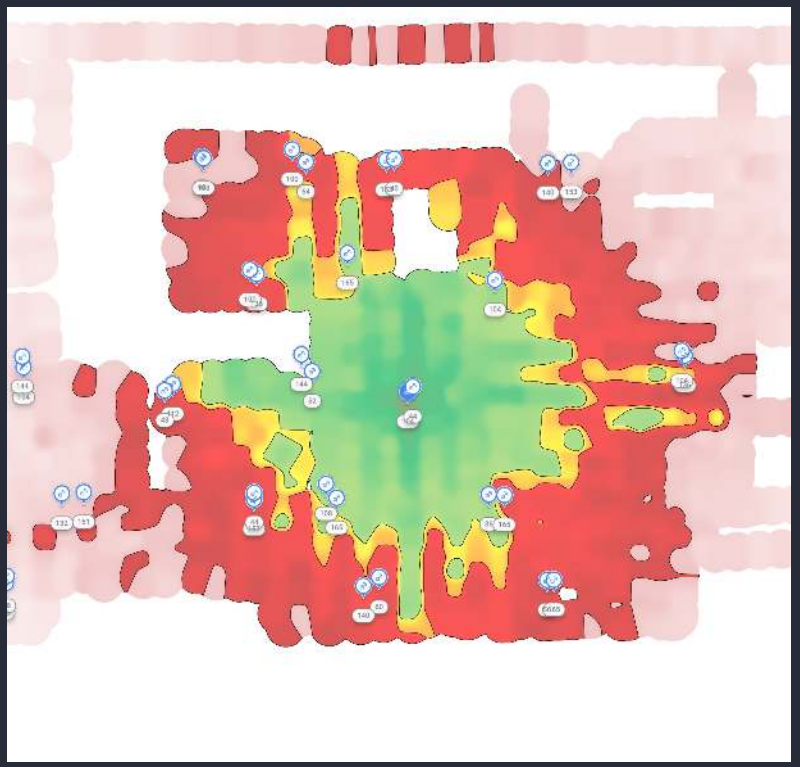
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Directional Antenna APs

Omni



Integrated Directional

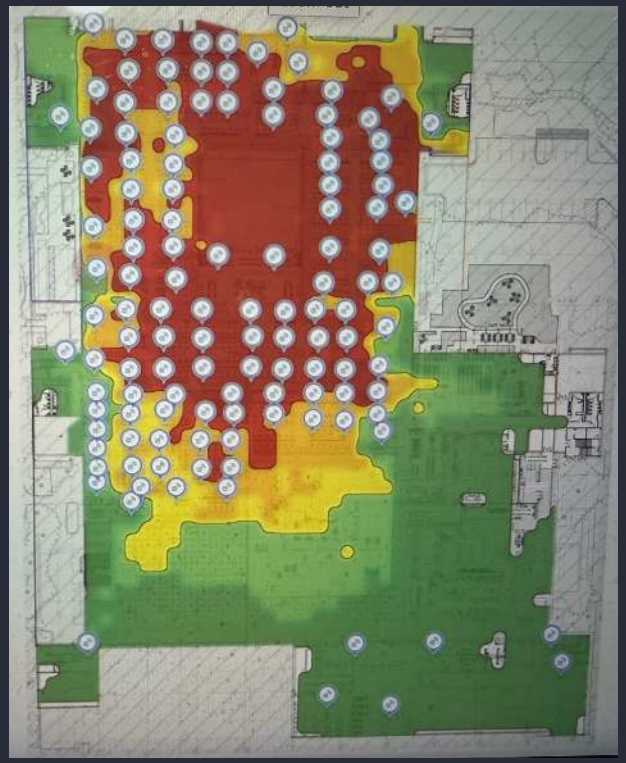


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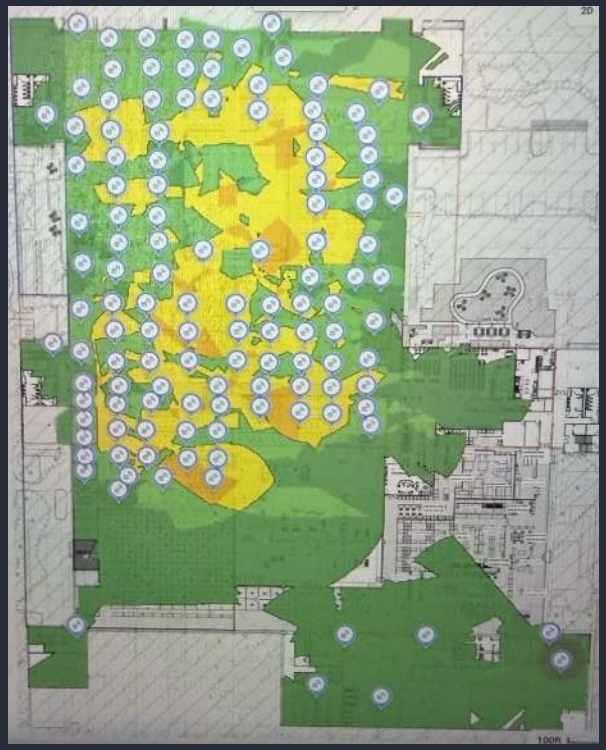
Why directional antenna APs in everyday deployments?

6 GHz Interference

Before - Omni



After - Integrated Directional

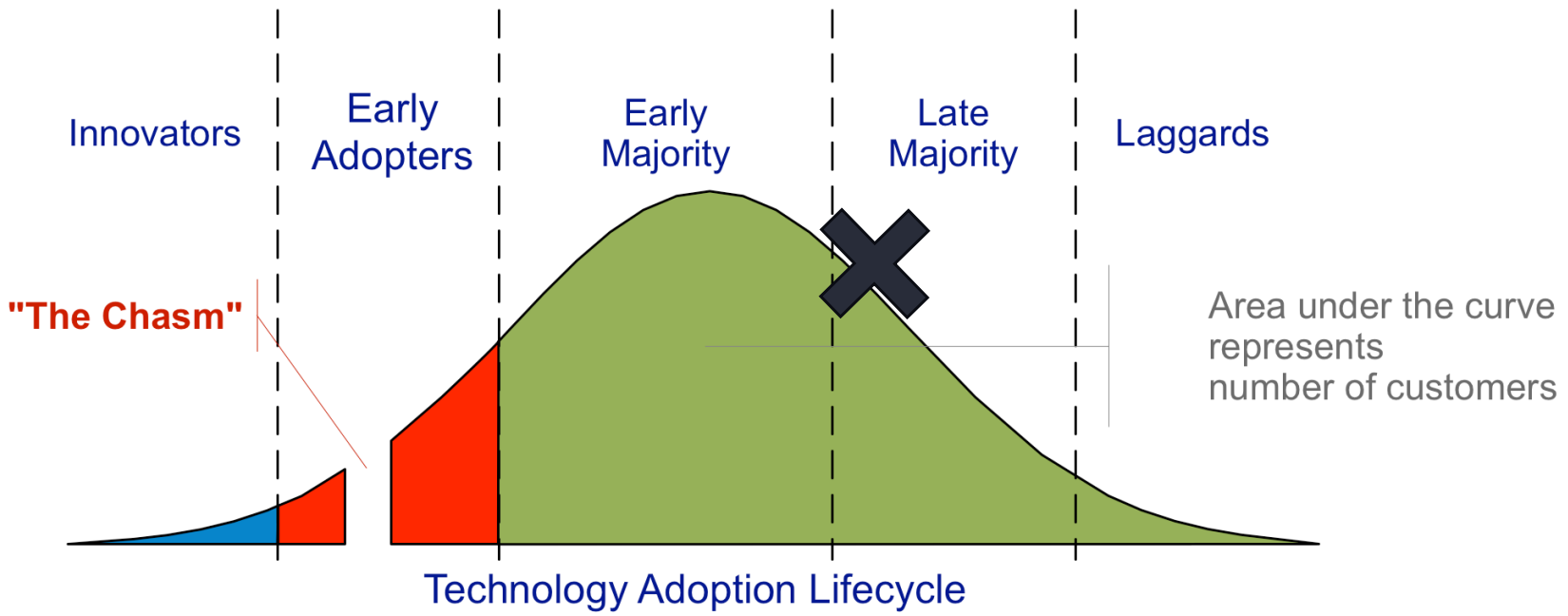


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Wi-Fi 7 Adoption

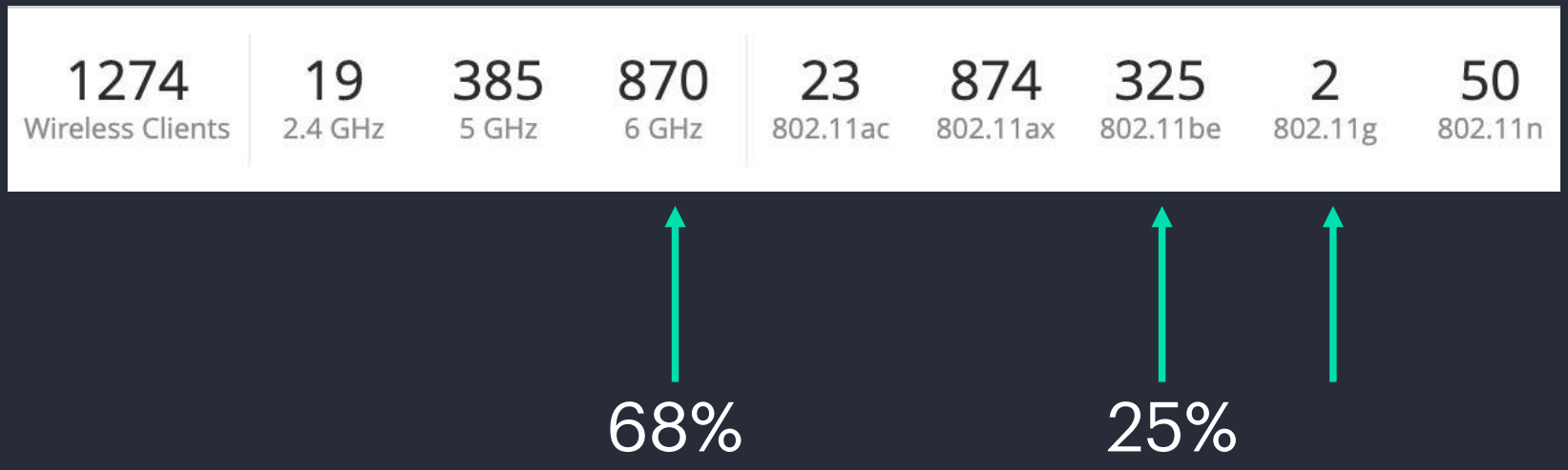
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Where are we with Wi-Fi 7?



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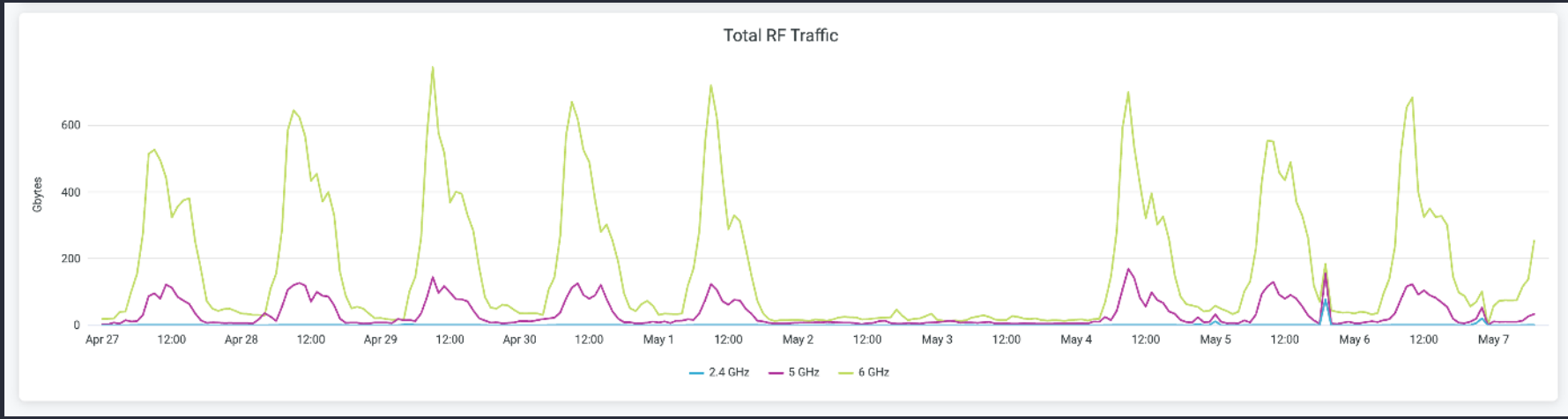
Wi-Fi 7 Adoption in Carpeted Enterprise



Juniper Business Use Only

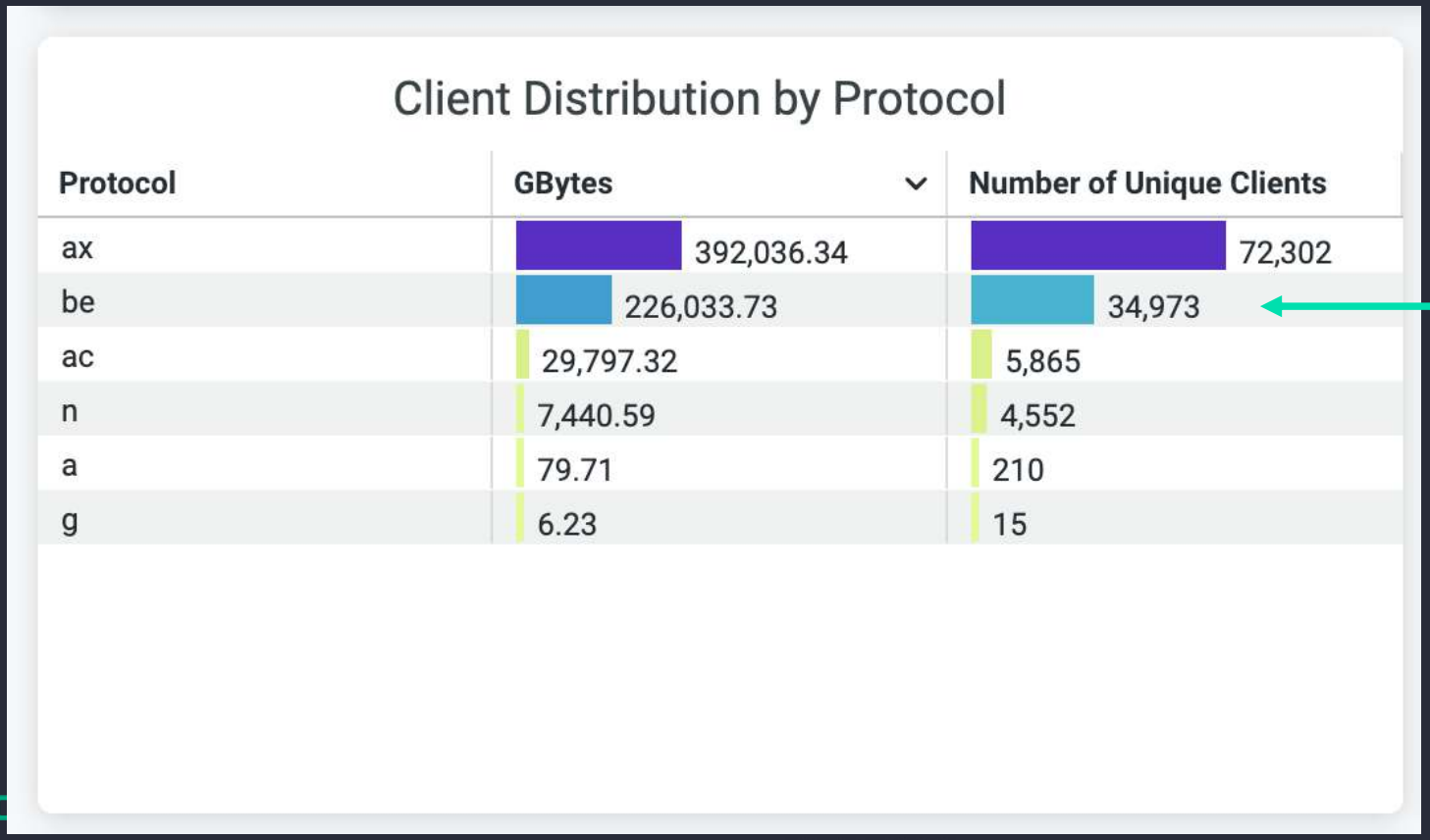
6 GHz Usage in Carpeted Enterprise

6 GHz usage = 3:1 5 GHz



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Wi-Fi 7 Adoption in Higher Ed



31%

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Wi-Fi 7 Adoption in Large Public Venue

Wi-Fi 7 **54% Clients**

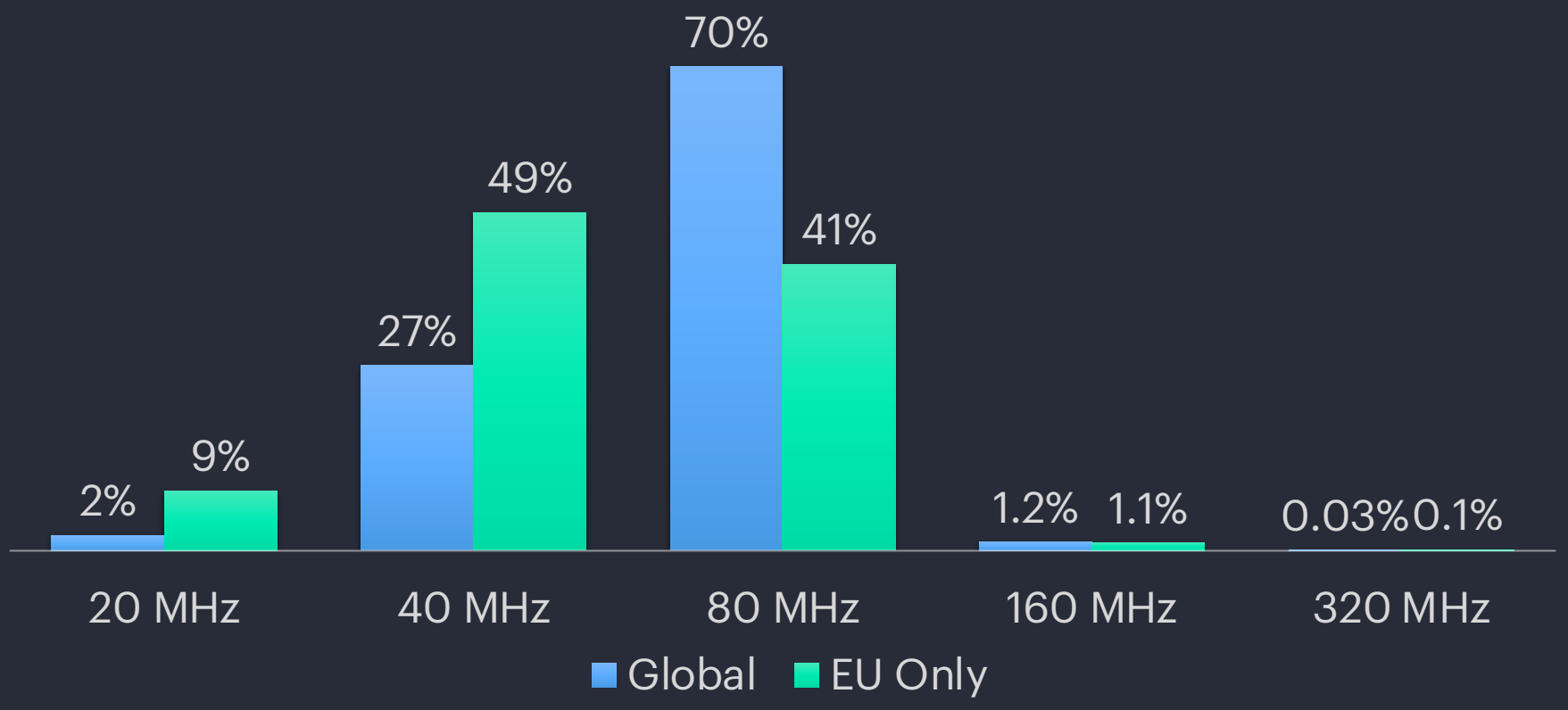
6 GHz **60% Clients**



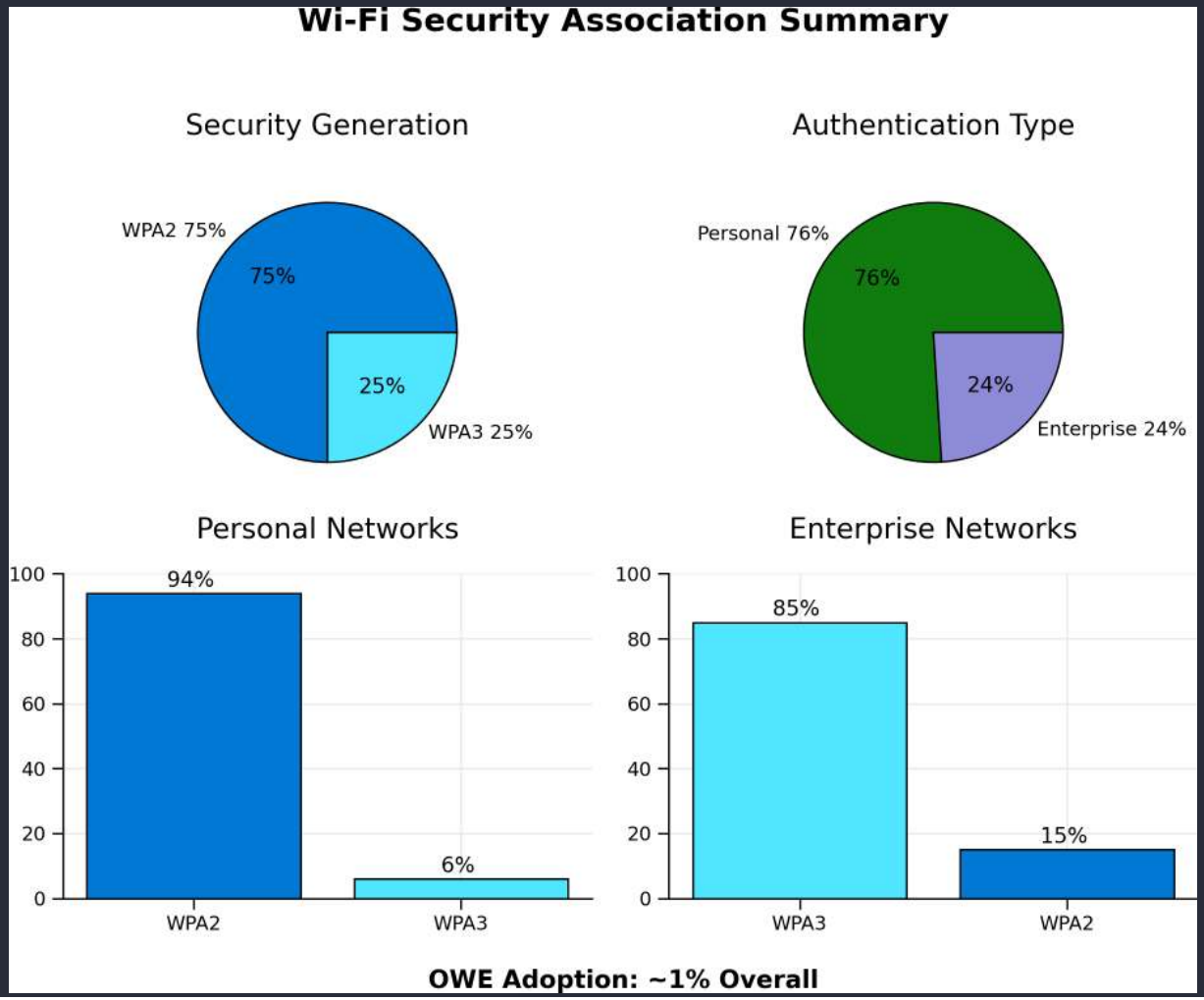
#WLPC

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6 GHz Channel Widths



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Wi-Fi Security Association Summary
Security Generation
WPA2 75%
75%
25%
WPA3 25%
Authentication Type
Personal 76%
76%
24%
Enterprise 24%
Personal Networks
100
80
60
40
20
0
94%
6%
WPA2
WPA3
Enterprise Networks
100
80
60
40
20
0
85%
15%
WPA3
WPA2
OWE Adoption: ~1% Overall

When OWE is Enabled – Clients Use it

OWE Transition and Pure OWE becoming common on Guest Networks

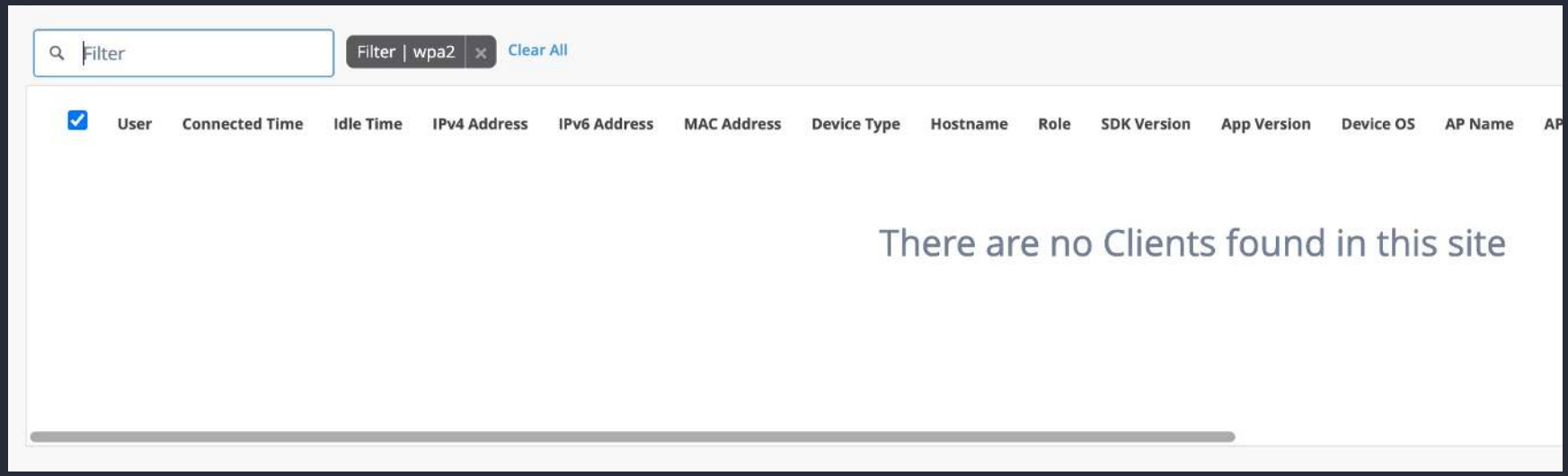
Client Mix on OWE Transition

95%
OWE

5%
Open

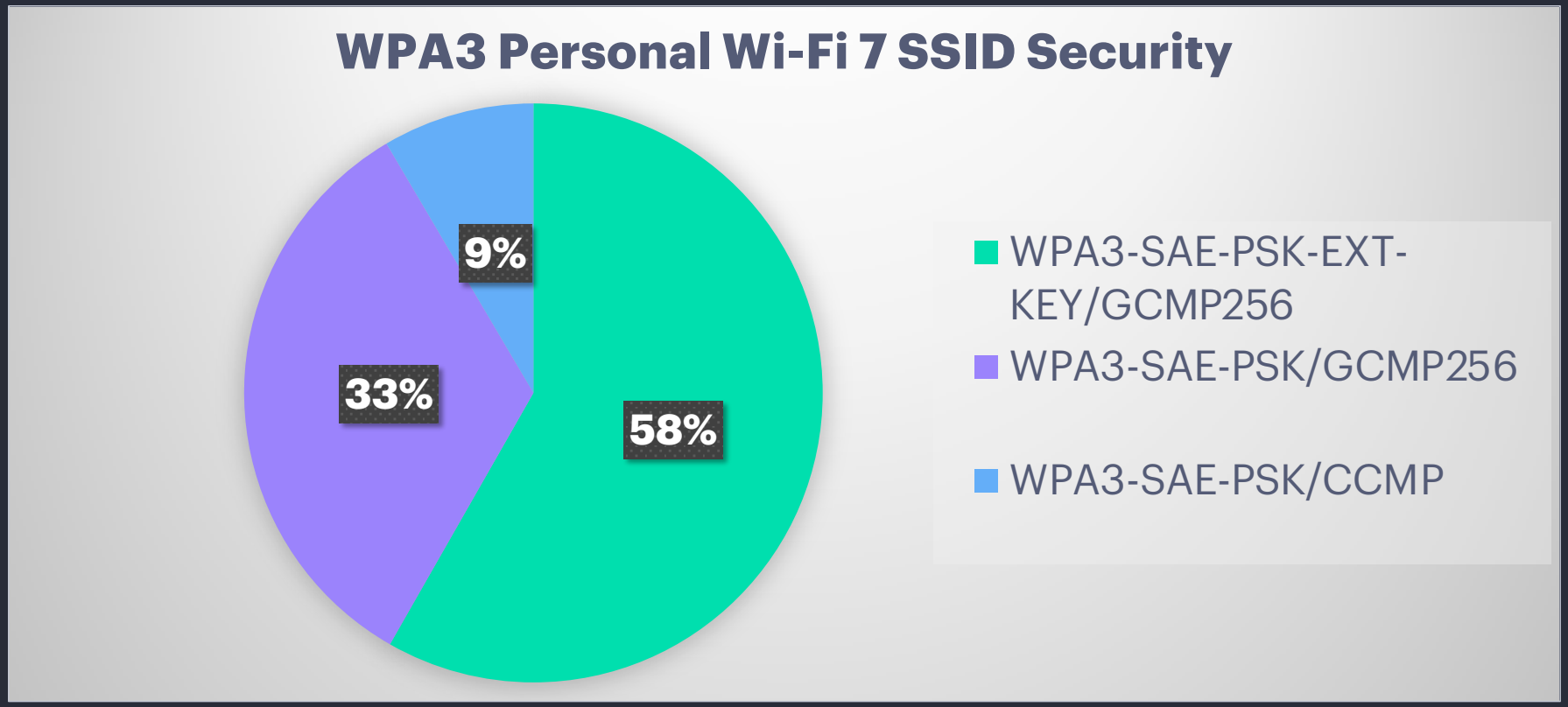
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WPA2 vs WPA3 on Public Passpoint



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What About Wi-Fi 7 Security?



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So...MLO?



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MLO Prerequisites

6 GHz (nice to have) – 75% of Wi-Fi 7 APs have 6 GHz enabled



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MLO Prerequisites

- **Security** – 1% using proper Wi-Fi 7 security.
- Does it matter?
  - It's anti-standard
  - But does not prevent the use of Wi-Fi 7 or MLO (vendor dependent)



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MLO Prerequisites

- **Interoperability**
  - Poor link selection with some clients
  - Inconsistent 11r support
  - Some clients don’t support MLO with 6 GHz
  - Windows recently added support for MLO with 802.1X



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MLO Adoption

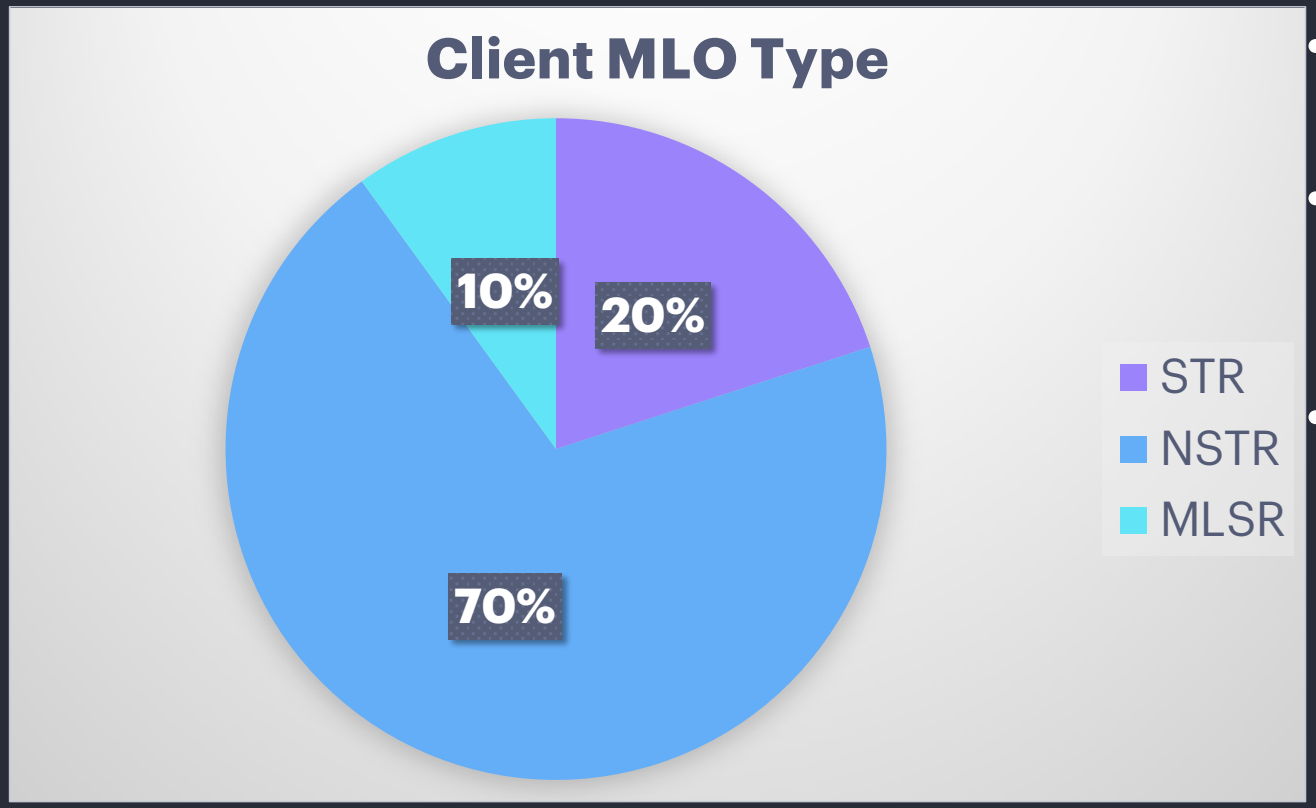
MLO

2%
Enabled



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MLO Type



- **STR** – High End Android
- **NSTR** – Generally Apple Devices
- **MLSR** – Generally Windows (Intel) and lower end Android

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Standard Power for UK

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Standard Power in the UK

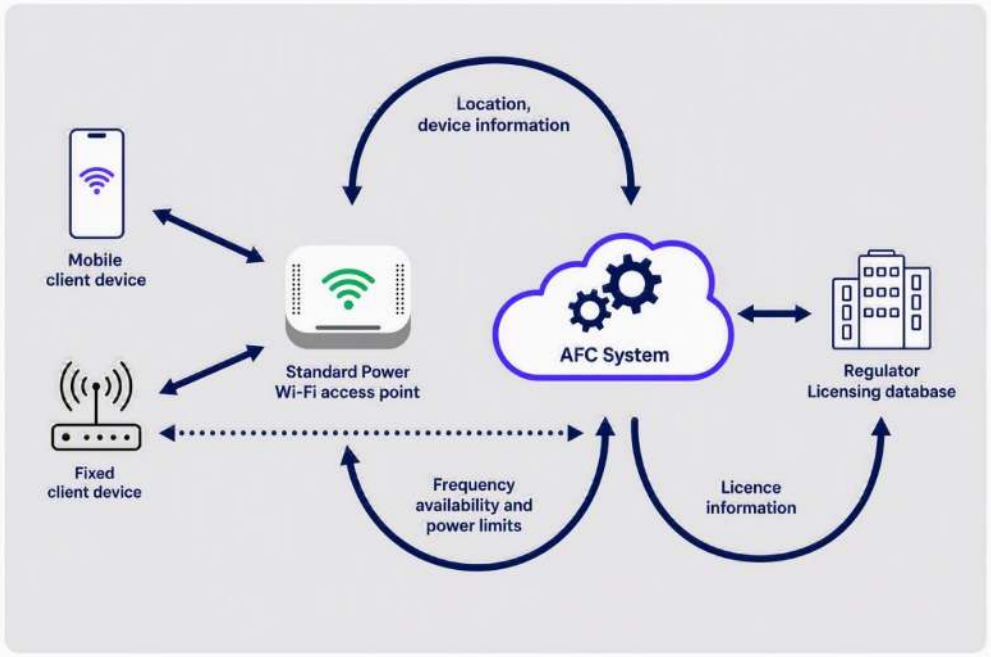
When?	Hopefully 2027 by the time AFCs are approved and AP/Clients certify under the new rules
More spectrum	Additional 160 MHz of “Wi-Fi Priority” (Wi-Fi has access)
	Additional 540 MHz of “Mobile Priority” (Wi-Fi may have access)
Higher Power	
Indoor / Outdoor use	

https://www.ofcom.org.uk/spectrum/innovative-use-of-spectrum/consultation-expanding-access-to-the-6-ghz-band-for-commercial-mobile-and-wi-fi-services
https://www.ofcom.org.uk/spectrum/innovative-use-of-spectrum/enabling-automated-frequency-coordination-afc-in-the-6-ghz-band

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How does AFC Work?

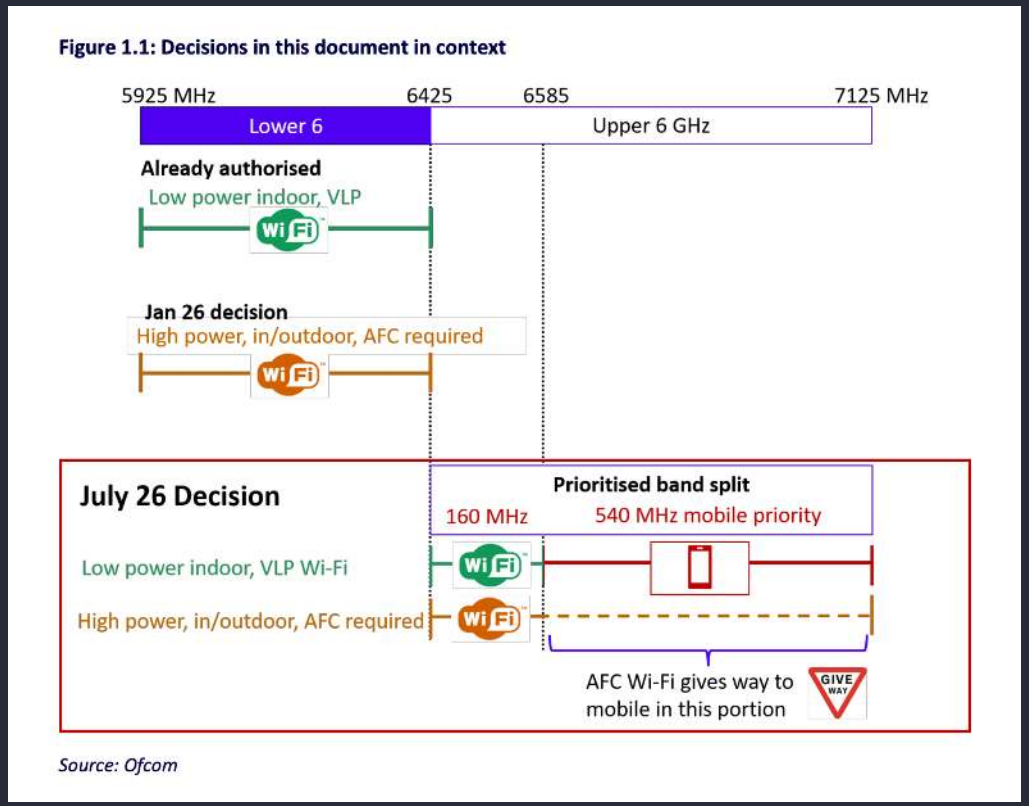
Figure 3: Overview of AFC operation



Source: Ofcom

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Here is the spectrum view



https://www.ofcom.org.uk/siteassets/resources/documents/consultations/category-3-4-weeks/consultation-expanding-access-to-the-6-ghz-band-for-commercial-mobile-and-wi-fi-services/main-document/statement-expanding-access-to-the-6-ghz-band-for-commercial-mobile-and-wi-fi-services.pdf?v=422807

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6GHz Regulatory Power Limits

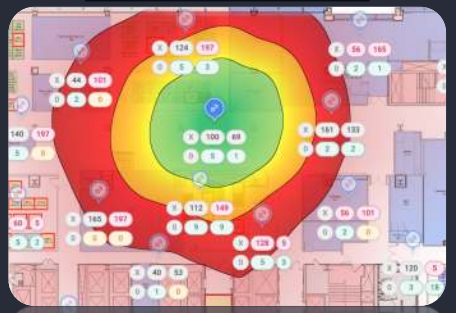
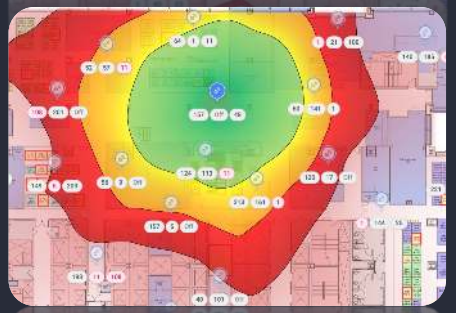
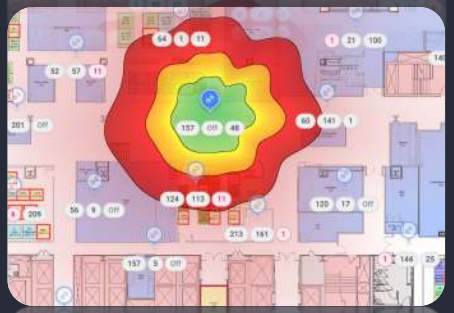
MQ TRAINING SERVICES

	US (FCC)		Europe (EU)		UK (Ofcom)	
	PSD	**EIRP**	**PSD**	**EIRP**	**PSD**	**EIRP**
Low Power Indoor Access Point	5 dBm/MHz	20 MHz: 18 dBm 40 MHz: 21 dBm 80 MHz: 24 dBm 160 MHz: 27 dBm **320 MHz: 30 dBm**	10 dBm/MHz	**23 dBm**	12.6dBm/MHz	**24 dBm**
Low Power Indoor Client	-1 dBm/MHz	20 MHz: 12 dBm 40 MHz: 15 dBm 80 MHz: 18 dBm 160 MHz: 21 dBm **320 MHz: 24 dBm** + 6 dBm below AP				
Standard Power Access Point	23 dBm/MHz	**36 dBm**	Not Allowed		**36 dBm**	
Standard Power Client	17 dBm/MHz	**30 dBm +** 6 dB below AP			**30 dBm +** 6 dB below AP	

© Peter Mackenzie

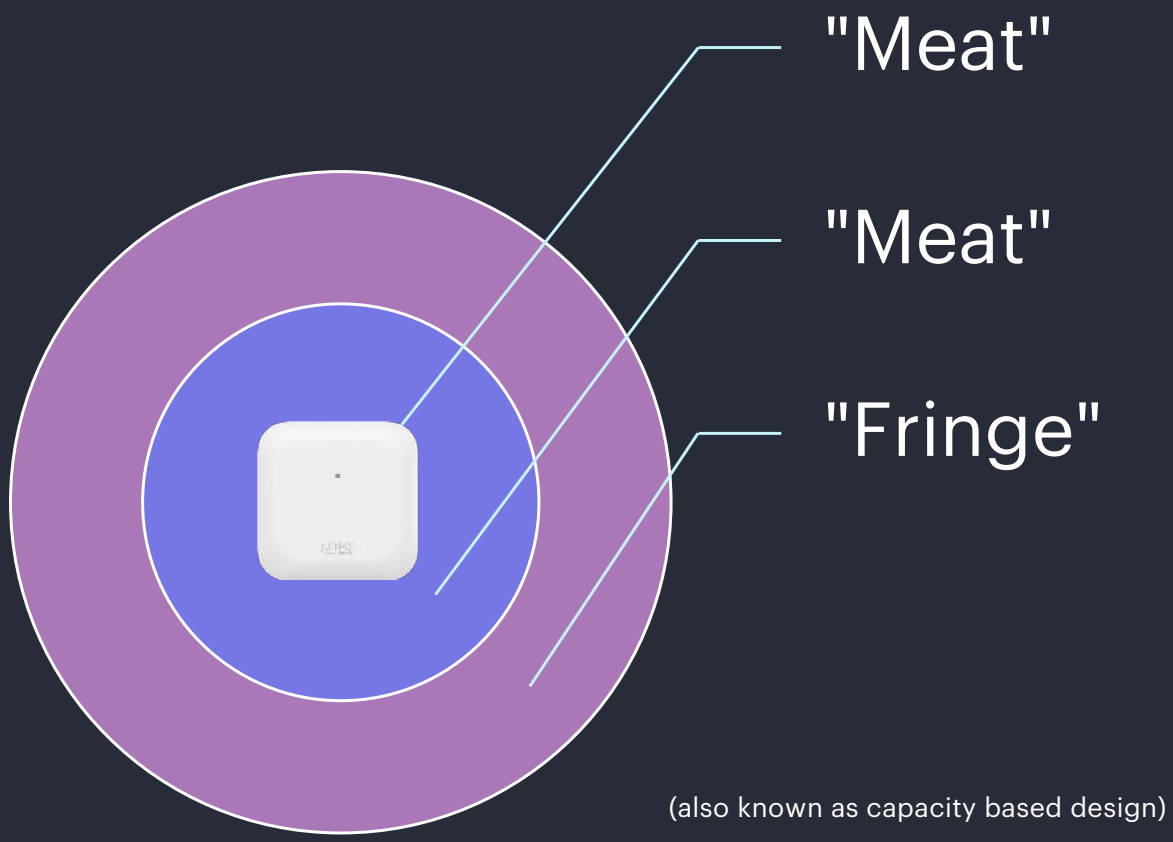
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Client Coverage in 6 GHz (US LPI)

	5 GHz	6 GHz
AP Coverage		
Client Coverage		

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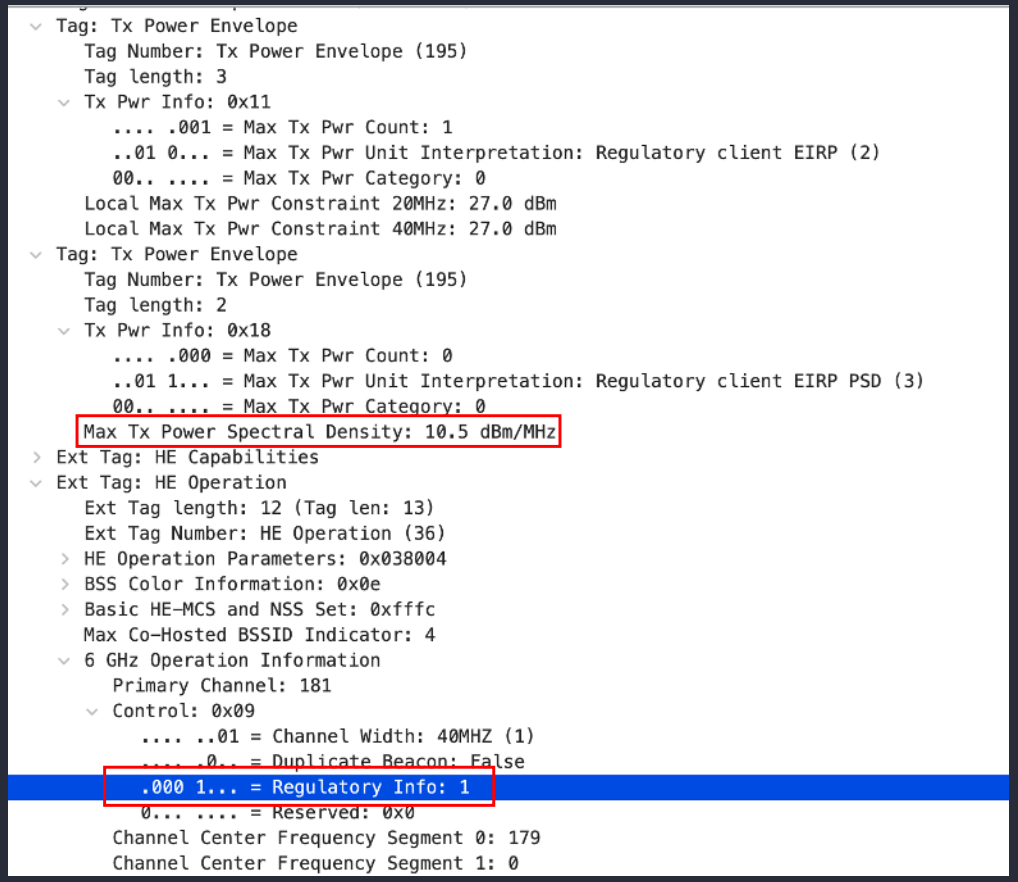
Scientific Method: Design 6 GHz for “Meat” of the Cell (US)



(also known as capacity based design)

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Information in the Beacon



Transmit Power Element IE – Lets the client know its allowed power

Regulatory Info Field – The AP's operating mode

0 = LPI
1 = SP

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MacBook transmit power (US)

18 dB Delta

20 MHz LPI

Test Frequency (MHz)	Maximum Conducted Output Power (dBm)					Gain (dBi)	EIRP (dBm)	EIRP Limit (dBm)	EIRP Margin (dB)
	A	B	C	D	∑				
5955	-3.45	-2.71	-	-	-0.05	6.40	6.35	24.00	-17.65
6175	-2.97	-1.76	-	-	0.68	6.10	6.78	24.00	-17.22
6415	-1.29	-1.46	-	-	1.64	5.40	7.04	24.00	-16.96

80 MHz LPI

Test Frequency (MHz)	Maximum Conducted Output Power (dBm)					Gain (dBi)	EIRP (dBm)	EIRP Limit (dBm)	EIRP Margin (dB)
	A	B	C	D	∑				
5985	2.67	2.90	-	-	5.80	6.40	12.20	24.00	-11.80
6145	3.08	3.59	-	-	6.35	6.10	12.45	24.00	-11.55
6385	4.17	4.15	-	-	7.17	5.40	12.57	24.00	-11.43

20 MHz SP

Test Frequency (MHz)	Maximum Conducted Output Power (dBm)					Gain (dBi)	EIRP (dBm)	EIRP Limit (dBm)	EIRP Margin (dB)
	A	B	C	D	∑				
5955	14.52	14.74	-	-	17.64	6.40	24.04	30.00	-5.96
6175	15.71	15.48	-	-	18.60	6.10	24.70	30.00	-5.30
6415	16.05	16.40	-	-	19.24	5.40	24.64	30.00	-5.36

Source: FCC

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Standard Power AP – Beacon Power



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LPI vs SP client

LPI

Protocol	Security	Band	RSSI
802.11ax	WPA3-EAP-FT/CCMP	6 GHz	-68 dBm

SP

Protocol	Security	Band	RSSI
802.11ax	WPA3-EAP-FT/CCMP	6 GHz	-59 dBm

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Does SP make a difference?

LPI



SP

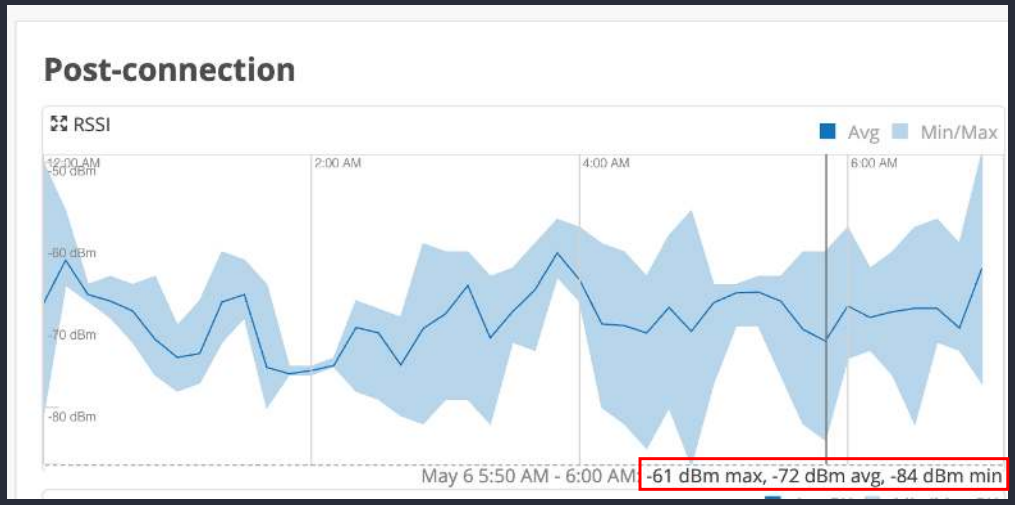


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Deployment Results

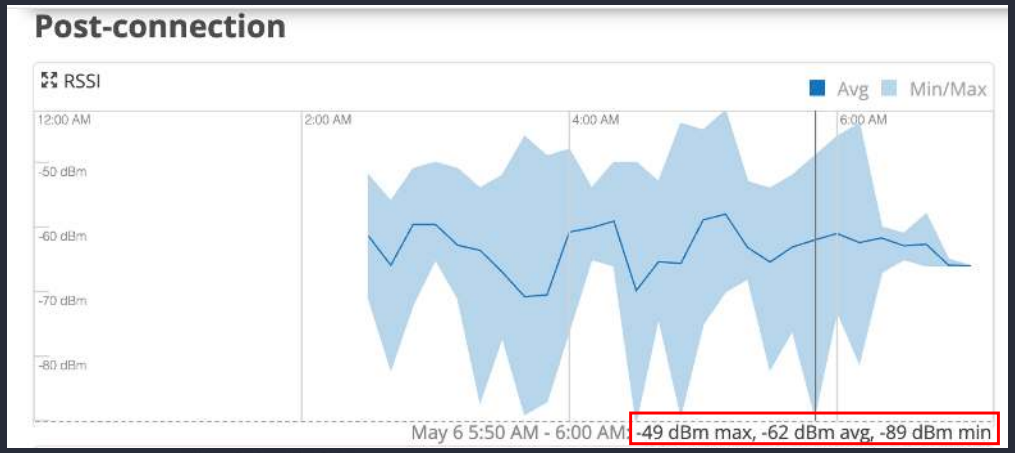
APs mounted at 20’ Height
Clients roaming

LPI



SP

Higher Average and Peak RSSI



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What is the verdict on Standard Power Indoors?

Pros: Indoor Standard Power

- Very useful in deployments with fringe 6 GHz coverage
- Seen to improve roaming and reduce interband roams where there may be fringe coverage, high ceilings, etc.

Cons: Indoor Standard Power

- Adds complexity (APs need geolocation)
- Reduces available channels compared to LPI

It’s a tool in the toolbox

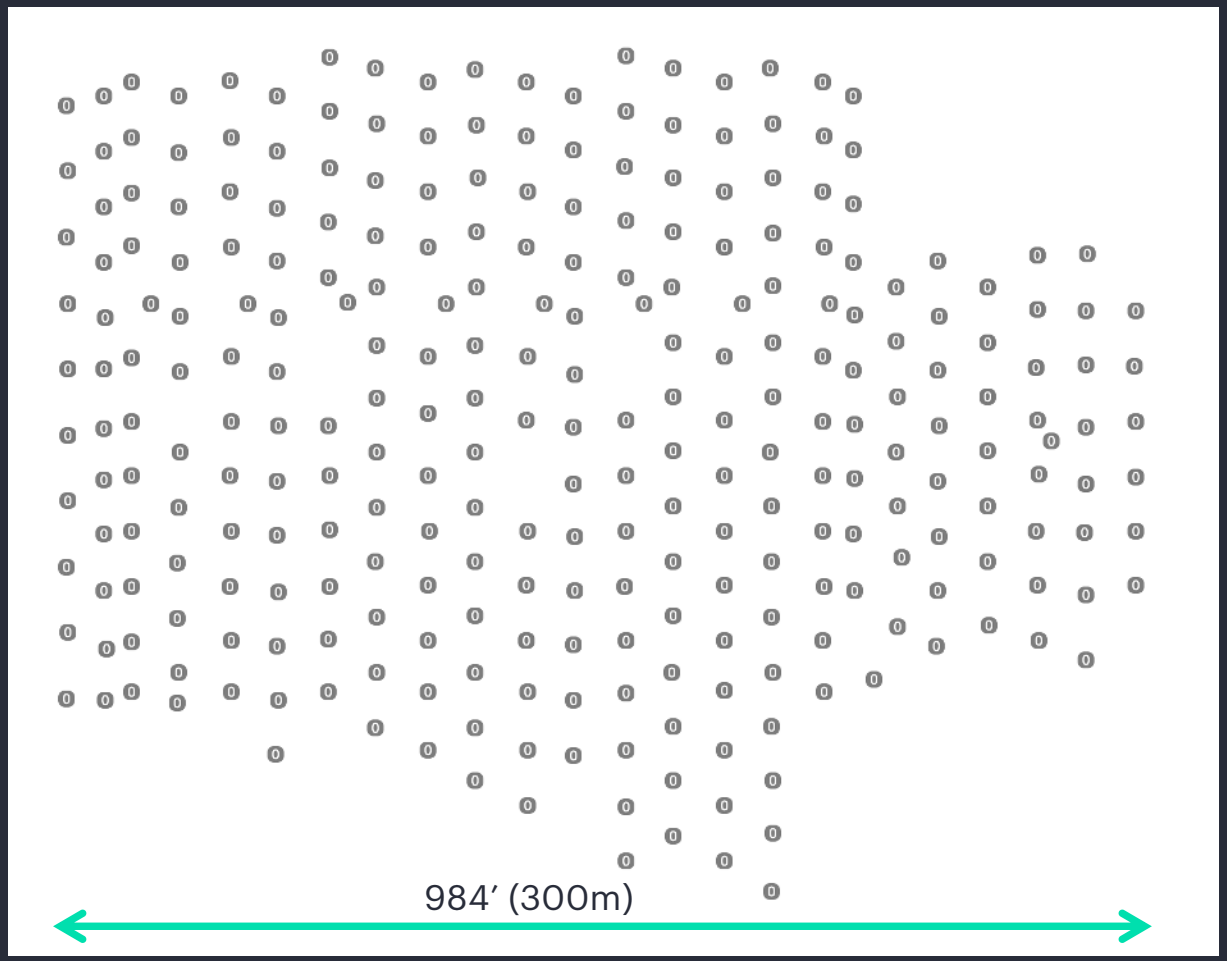
In the US it allows for higher client Tx power
In the UK it allows for more spectrum



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278 APs

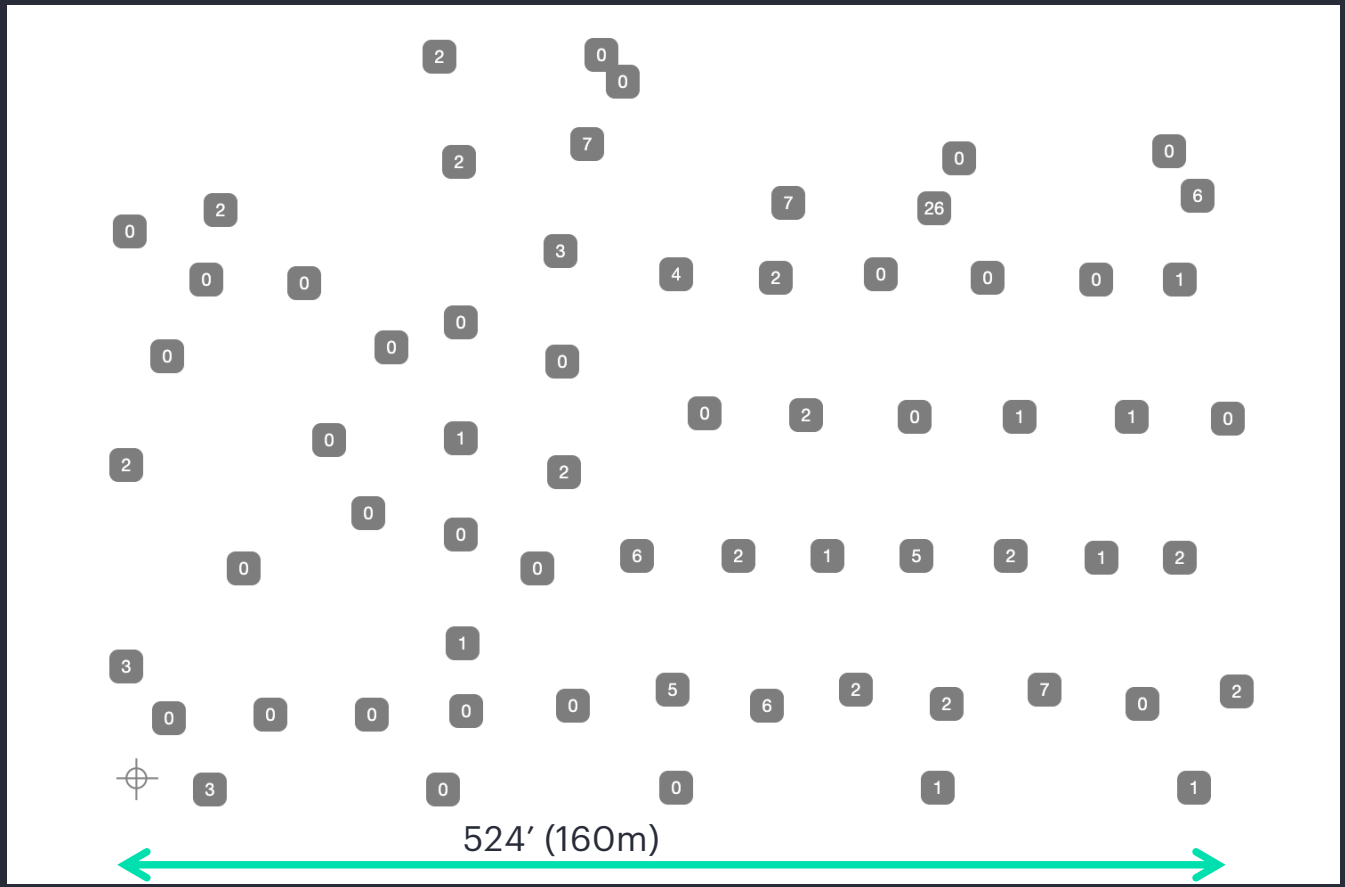
152 GPS
126 Derived



Juniper Business Use Only

41 APs

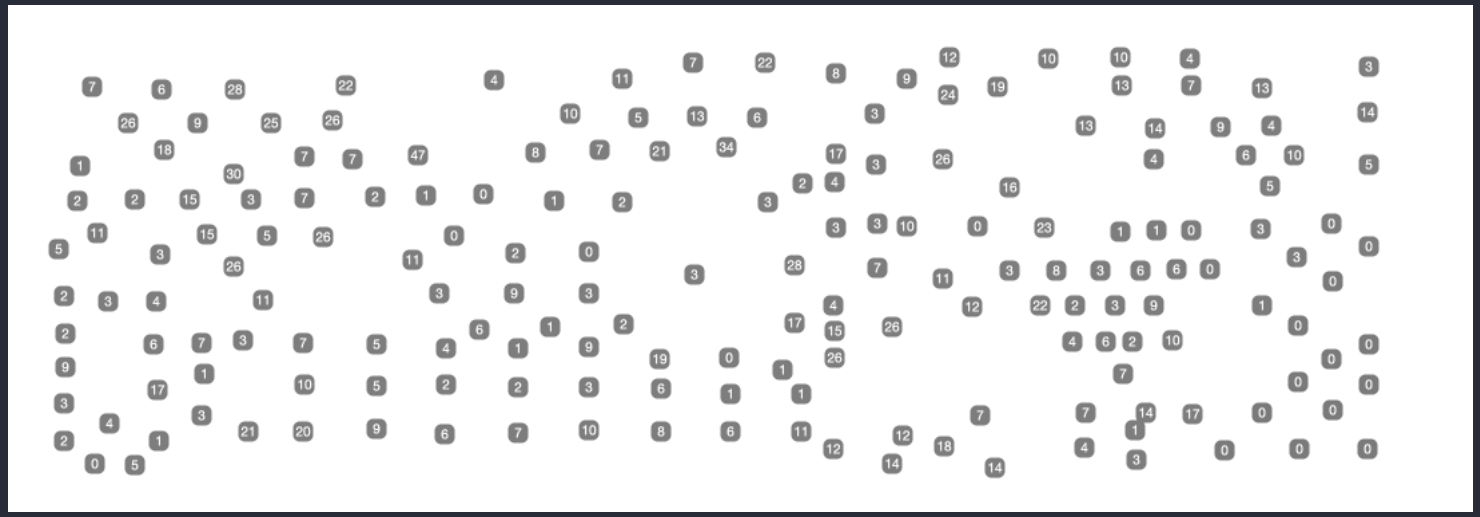
21 GPS
20 Derived



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231 APs

4 GPS
227 Derived



1735′ (529m)

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Thank You



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